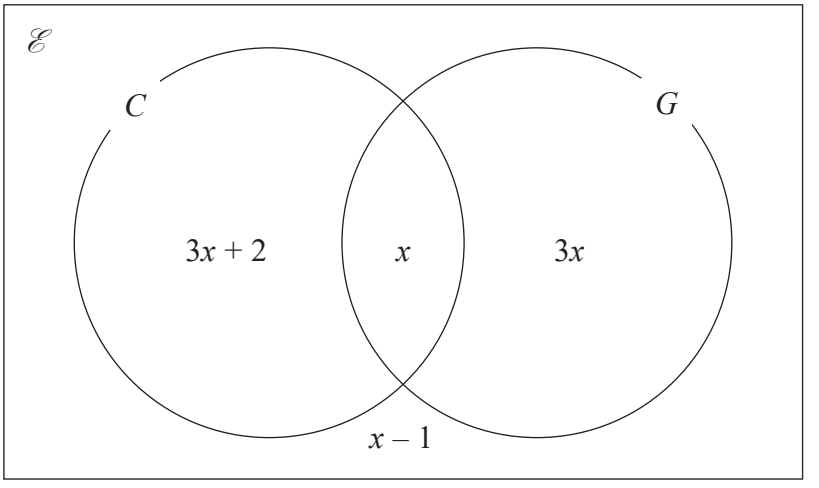
23 The Venn diagram gives some information about the number of people in a room.

$\mathscr{E}$ = {all people in the room}
C = {children}
G = {people wearing glasses}



Two children are picked at random from all the children in the room.

The probability that both children wear glasses is $\frac{2}{51}$

Work out the total number of people in the room.
You must show all your working.

Children: $(3x + 2) + x = 4x + 2$; glasses: x

$$\frac{x}{4x+2} \times \frac{x-1}{4x+1} = \frac{2}{51}$$

2 marks total
Two picks WITHOUT replacement.

$51x(x - 1) = 2(4x + 2)(4x + 1)$

$19x^2 - 75x - 4 = 0$

1 mark
Expand and collect all terms.

$(19x + 1)(x - 4) = 0$

1 mark
Factorise and select a valid count.

$x = 4$; reject $x = -1/19$ (counts cannot be negative).

Total $= (3x+2) + x + 3x + (x-1) = 8x+1 = 33$

1 mark
Add ALL four Venn regions.

33

(Total for Question 23 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA